

ASTHA RAVEN - Database Schema

ASTHA RAVEN - Estrutura do Banco de Dados

Tabela: aluno

```
`sql
CREATE TABLE `aluno` (
  `ra` int NOT NULL,
  `nome` varchar(255) DEFAULT NULL,
  `id` int NOT NULL,
  `email` varchar(255) DEFAULT NULL,
  PRIMARY KEY (`ra`)
) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4 COLLATE=utf8mb4_0900_ai_ci;
```

Tabela: audit_log

```
`sql
CREATE TABLE `audit_log` (
  `id` bigint NOT NULL AUTO_INCREMENT,
  `user_id` int DEFAULT NULL,
  `action` varchar(50) CHARACTER SET utf8mb4 COLLATE utf8mb4_unicode_ci NOT
  NULL COMMENT 'CREATE, UPDATE, DELETE, LOGIN, LOGOUT, EXPORT',
  `entity_type` varchar(50) CHARACTER SET utf8mb4 COLLATE utf8mb4_unicode_ci
  DEFAULT NULL,
  `entity_id` int DEFAULT NULL,
  `old_values` json DEFAULT NULL,
  `new_values` json DEFAULT NULL,
  `ip_address` varchar(45) CHARACTER SET utf8mb4 COLLATE utf8mb4_unicode_ci
  DEFAULT NULL,
  `user_agent` varchar(500) CHARACTER SET utf8mb4 COLLATE utf8mb4_unicode_ci
  DEFAULT NULL,
  `created_at` timestamp NULL DEFAULT CURRENT_TIMESTAMP,
  PRIMARY KEY (`id`),
  KEY `idx_user_action` (`user_id`,`action`),
  KEY `idx_entity` (`entity_type`,`entity_id`),
  KEY `idx_date` (`created_at`),
  CONSTRAINT `audit_log_ibfk_1` FOREIGN KEY (`user_id`) REFERENCES `users`
  (`id`) ON DELETE SET NULL
) ENGINE=InnoDB AUTO_INCREMENT=17 DEFAULT CHARSET=utf8mb4
  COLLATE=utf8mb4_unicode_ci;
```

Tabela: autor

```
`sql
CREATE TABLE `autor` (
  `id` int NOT NULL,
  `labubu_velociraptor_email` varchar(255) DEFAULT NULL,
  `nacionalidade` varchar(255) DEFAULT NULL,
  `nome` varchar(255) DEFAULT NULL,
  PRIMARY KEY (`id`)
) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4 COLLATE=utf8mb4_0900_ai_ci;
```

Tabela: categories

```
`sql
CREATE TABLE `categories` (
  `id` int NOT NULL AUTO_INCREMENT,
  `name` varchar(100) CHARACTER SET utf8mb4 COLLATE utf8mb4_unicode_ci NOT
  NULL,
```

```

    `slug` varchar(100) CHARACTER SET utf8mb4 COLLATE utf8mb4_unicode_ci NOT
NULL,
    `description` text CHARACTER SET utf8mb4 COLLATE utf8mb4_unicode_ci,
    `color` varchar(7) CHARACTER SET utf8mb4 COLLATE utf8mb4_unicode_ci DEFAULT
'#6C63FF',
    `parent_id` int DEFAULT NULL,
    `created_at` timestamp NULL DEFAULT CURRENT_TIMESTAMP,
    PRIMARY KEY (`id`),
    UNIQUE KEY `slug` (`slug`),
    KEY `parent_id` (`parent_id`),
    CONSTRAINT `categories_ibfk_1` FOREIGN KEY (`parent_id`) REFERENCES
`categories` (`id`) ON DELETE SET NULL
) ENGINE=InnoDB AUTO_INCREMENT=6 DEFAULT CHARSET=utf8mb4
COLLATE=utf8mb4_unicode_ci;

```

Tabela: component_materials

```

`sql
CREATE TABLE `component_materials` (
    `id` int NOT NULL AUTO_INCREMENT,
    `component_id` int NOT NULL,
    `material_id` int NOT NULL,
    `quantity` int NOT NULL DEFAULT '1',
    PRIMARY KEY (`id`),
    UNIQUE KEY `uk_comp_mat` (`component_id`,`material_id`),
    KEY `material_id` (`material_id`),
    CONSTRAINT `component_materials_ibfk_1` FOREIGN KEY (`component_id`)
REFERENCES `components` (`id`) ON DELETE CASCADE,
    CONSTRAINT `component_materials_ibfk_2` FOREIGN KEY (`material_id`)
REFERENCES `materials` (`id`) ON DELETE CASCADE
) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4 COLLATE=utf8mb4_unicode_ci;

```

Tabela: components

```

`sql
CREATE TABLE `components` (
    `id` int NOT NULL AUTO_INCREMENT,
    `name` varchar(200) CHARACTER SET utf8mb4 COLLATE utf8mb4_unicode_ci NOT
NULL,
    `description` text CHARACTER SET utf8mb4 COLLATE utf8mb4_unicode_ci,
    `created_by` int DEFAULT NULL,
    `created_at` timestamp NULL DEFAULT CURRENT_TIMESTAMP,
    `updated_at` timestamp NULL DEFAULT CURRENT_TIMESTAMP ON UPDATE
CURRENT_TIMESTAMP,
    PRIMARY KEY (`id`),
    KEY `created_by` (`created_by`),
    CONSTRAINT `components_ibfk_1` FOREIGN KEY (`created_by`) REFERENCES
`users` (`id`) ON DELETE SET NULL
) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4 COLLATE=utf8mb4_unicode_ci;

```

Tabela: contact_messages

```

`sql
CREATE TABLE `contact_messages` (
    `id` int NOT NULL AUTO_INCREMENT,
    `name` varchar(100) CHARACTER SET utf8mb4 COLLATE utf8mb4_unicode_ci NOT
NULL,
    `email` varchar(255) CHARACTER SET utf8mb4 COLLATE utf8mb4_unicode_ci NOT
NULL,
    `subject` varchar(200) CHARACTER SET utf8mb4 COLLATE utf8mb4_unicode_ci
DEFAULT 'Geral',

```

```

    `message` text CHARACTER SET utf8mb4 COLLATE utf8mb4_unicode_ci NOT NULL,
    `is_read` tinyint(1) NOT NULL DEFAULT '0',
    `replied_at` timestamp NULL DEFAULT NULL,
    `created_at` timestamp NULL DEFAULT CURRENT_TIMESTAMP,
    PRIMARY KEY (`id`),
    KEY `idx_email` (`email`),
    KEY `idx_date` (`created_at`)
) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4 COLLATE=utf8mb4_unicode_ci;

```

Tabela: curso

```

`sql
CREATE TABLE `curso` (
  `id` int NOT NULL,
  `duracao_horas` double DEFAULT NULL,
  `nivel` int DEFAULT NULL,
  `nome` varchar(255) DEFAULT NULL,
  PRIMARY KEY (`id`)
) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4 COLLATE=utf8mb4_0900_ai_ci;

```

Tabela: disciplina

```

`sql
CREATE TABLE `disciplina` (
  `carga_horaria` int DEFAULT NULL,
  `codigo` int NOT NULL,
  `nome` varchar(255) DEFAULT NULL,
  `sigla` varchar(255) DEFAULT NULL,
  PRIMARY KEY (`codigo`)
) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4 COLLATE=utf8mb4_0900_ai_ci;

```

Tabela: filme

```

`sql
CREATE TABLE `filme` (
  `ano_lancamento` int DEFAULT NULL,
  `codigo` int NOT NULL,
  `nota` double DEFAULT NULL,
  `diretor` varchar(255) DEFAULT NULL,
  `genero` varchar(255) DEFAULT NULL,
  `titulo` varchar(255) DEFAULT NULL,
  PRIMARY KEY (`codigo`)
) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4 COLLATE=utf8mb4_0900_ai_ci;

```

Tabela: insumo

```

`sql
CREATE TABLE `insumo` (
  `id` int NOT NULL AUTO_INCREMENT,
  `tipo` varchar(50) NOT NULL,
  `nome` varchar(100) NOT NULL,
  `valor` decimal(10,2) NOT NULL,
  `data_aquisicao` date NOT NULL,
  PRIMARY KEY (`id`)
) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4 COLLATE=utf8mb4_0900_ai_ci;

```

Tabela: invoices

```

`sql
CREATE TABLE `invoices` (
  `id` int NOT NULL AUTO_INCREMENT,

```

```

    `subscription_id` int NOT NULL,
    `amount` decimal(10,2) NOT NULL,
    `currency` varchar(5) CHARACTER SET utf8mb4 COLLATE utf8mb4_unicode_ci
    DEFAULT 'BRL',
    `status` enum('pending','paid','failed','refunded') CHARACTER SET utf8mb4
    COLLATE utf8mb4_unicode_ci DEFAULT 'pending',
    `paid_at` timestamp NULL DEFAULT NULL,
    `due_date` date NOT NULL,
    `created_at` timestamp NULL DEFAULT CURRENT_TIMESTAMP,
    PRIMARY KEY (`id`),
    KEY `subscription_id` (`subscription_id`),
    CONSTRAINT `invoices_ibfk_1` FOREIGN KEY (`subscription_id`) REFERENCES
    `subscriptions` (`id`) ON DELETE CASCADE
) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4 COLLATE=utf8mb4_unicode_ci;

```

Tabela: livro

```

`sql
CREATE TABLE `livro` (
    `codigo` int NOT NULL,
    `ano_publicacao` int DEFAULT NULL,
    `autor` varchar(255) DEFAULT NULL,
    `genero` varchar(255) DEFAULT NULL,
    `titulo` varchar(255) DEFAULT NULL,
    PRIMARY KEY (`codigo`)
) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4 COLLATE=utf8mb4_0900_ai_ci;

```

Tabela: material_files

```

`sql
CREATE TABLE `material_files` (
    `id` int NOT NULL AUTO_INCREMENT,
    `material_id` int NOT NULL,
    `category` varchar(50) CHARACTER SET utf8mb4 COLLATE utf8mb4_unicode_ci NOT
    NULL,
    `filename` varchar(300) CHARACTER SET utf8mb4 COLLATE utf8mb4_unicode_ci
    NOT NULL,
    `mime_type` varchar(100) CHARACTER SET utf8mb4 COLLATE utf8mb4_unicode_ci
    DEFAULT NULL,
    `file_data` longtext CHARACTER SET utf8mb4 COLLATE utf8mb4_unicode_ci,
    `file_url` varchar(500) CHARACTER SET utf8mb4 COLLATE utf8mb4_unicode_ci
    DEFAULT NULL,
    `description` text CHARACTER SET utf8mb4 COLLATE utf8mb4_unicode_ci,
    `file_size_kb` int DEFAULT NULL,
    `created_at` timestamp NULL DEFAULT CURRENT_TIMESTAMP,
    PRIMARY KEY (`id`),
    KEY `idx_material` (`material_id`),
    CONSTRAINT `material_files_ibfk_1` FOREIGN KEY (`material_id`) REFERENCES
    `materials` (`id`) ON DELETE CASCADE
) ENGINE=InnoDB AUTO_INCREMENT=3 DEFAULT CHARSET=utf8mb4
    COLLATE=utf8mb4_unicode_ci;

```

Tabela: materials

```

`sql
CREATE TABLE `materials` (
    `id` int NOT NULL AUTO_INCREMENT,
    `code` varchar(20) CHARACTER SET utf8mb4 COLLATE utf8mb4_unicode_ci NOT
    NULL COMMENT 'Ex: MERB-00000001',
    `description` varchar(300) CHARACTER SET utf8mb4 COLLATE utf8mb4_unicode_ci
    NOT NULL,

```

```

`brand` varchar(100) CHARACTER SET utf8mb4 COLLATE utf8mb4_unicode_ci
DEFAULT NULL,
`model` varchar(100) CHARACTER SET utf8mb4 COLLATE utf8mb4_unicode_ci
DEFAULT NULL,
`category_id` int DEFAULT NULL,
`sector` varchar(100) CHARACTER SET utf8mb4 COLLATE utf8mb4_unicode_ci
DEFAULT NULL COMMENT 'Ex: Assembly, Paint, Body',
`location` varchar(50) CHARACTER SET utf8mb4 COLLATE utf8mb4_unicode_ci
DEFAULT NULL COMMENT 'Ex: A1-N2-P3',
`quantity` int NOT NULL DEFAULT '0',
`min_qty` int NOT NULL DEFAULT '0',
`max_qty` int NOT NULL DEFAULT '0',
`unit_price` decimal(12,2) DEFAULT '0.00',
`currency` varchar(5) CHARACTER SET utf8mb4 COLLATE utf8mb4_unicode_ci
DEFAULT 'BRL',
`unit` varchar(20) CHARACTER SET utf8mb4 COLLATE utf8mb4_unicode_ci DEFAULT
'UN' COMMENT 'UN, KG, L, M, CX',
`barcode` varchar(50) CHARACTER SET utf8mb4 COLLATE utf8mb4_unicode_ci
DEFAULT NULL,
`image_url` varchar(500) CHARACTER SET utf8mb4 COLLATE utf8mb4_unicode_ci
DEFAULT NULL,
`image_data` longtext CHARACTER SET utf8mb4 COLLATE utf8mb4_unicode_ci
COMMENT 'Base64 para uso offline',
`manual_url` varchar(500) CHARACTER SET utf8mb4 COLLATE utf8mb4_unicode_ci
DEFAULT NULL,
`manual_data` longtext CHARACTER SET utf8mb4 COLLATE utf8mb4_unicode_ci
COMMENT 'PDF em Base64',
`manual_filename` varchar(200) CHARACTER SET utf8mb4 COLLATE
utf8mb4_unicode_ci DEFAULT NULL,
`interchangeable` tinyint(1) NOT NULL DEFAULT '0',
`interchange_with` text CHARACTER SET utf8mb4 COLLATE utf8mb4_unicode_ci
COMMENT 'Códigos substitutos separados por vírgula',
`notes` text CHARACTER SET utf8mb4 COLLATE utf8mb4_unicode_ci,
`machinery` varchar(200) CHARACTER SET utf8mb4 COLLATE utf8mb4_unicode_ci
DEFAULT NULL,
`is_active` tinyint(1) NOT NULL DEFAULT '1',
`created_by` int DEFAULT NULL,
`created_at` timestamp NULL DEFAULT CURRENT_TIMESTAMP,
`updated_at` timestamp NULL DEFAULT CURRENT_TIMESTAMP ON UPDATE
CURRENT_TIMESTAMP,
`org_id` varchar(50) CHARACTER SET utf8mb4 COLLATE utf8mb4_unicode_ci
DEFAULT NULL COMMENT 'Organização dona do material',
`machinery_image` longtext CHARACTER SET utf8mb4 COLLATE utf8mb4_unicode_ci
COMMENT 'Imagem da máquina em Base64',
`machinery_desc` text CHARACTER SET utf8mb4 COLLATE utf8mb4_unicode_ci
COMMENT 'Descrição da máquina',
`usage_desc` text CHARACTER SET utf8mb4 COLLATE utf8mb4_unicode_ci COMMENT
'Descrição de uso do material',
`manufacturer_url` varchar(500) CHARACTER SET utf8mb4 COLLATE
utf8mb4_unicode_ci DEFAULT NULL COMMENT 'Site do fabricante',
`warehouse_id` int DEFAULT NULL COMMENT 'ID do armazém (futuro)',
PRIMARY KEY (`id`),
UNIQUE KEY `code` (`code`),
KEY `idx_code` (`code`),
KEY `idx_sector` (`sector`),
KEY `idx_brand` (`brand`),
KEY `idx_barcode` (`barcode`),
KEY `idx_category` (`category_id`),
KEY `created_by` (`created_by`),
CONSTRAINT `materials_ibfk_1` FOREIGN KEY (`category_id`) REFERENCES
`categories` (`id`) ON DELETE SET NULL,

```

```

        CONSTRAINT `materials_ibfk_2` FOREIGN KEY (`created_by`) REFERENCES `users`
        (`id`) ON DELETE SET NULL
    ) ENGINE=InnoDB AUTO_INCREMENT=17 DEFAULT CHARSET=utf8mb4
    COLLATE=utf8mb4_unicode_ci;
`

```

Tabela: movements

```

`sql
CREATE TABLE `movements` (
    `id` int NOT NULL AUTO_INCREMENT,
    `material_id` int NOT NULL,
    `type` enum('entrada','saida') CHARACTER SET utf8mb4 COLLATE
utf8mb4_unicode_ci NOT NULL,
    `quantity` int NOT NULL,
    `nf_number` varchar(50) CHARACTER SET utf8mb4 COLLATE utf8mb4_unicode_ci
DEFAULT NULL,
    `responsible` varchar(100) CHARACTER SET utf8mb4 COLLATE utf8mb4_unicode_ci
DEFAULT NULL,
    `reason` varchar(300) CHARACTER SET utf8mb4 COLLATE utf8mb4_unicode_ci
DEFAULT NULL,
    `notes` text CHARACTER SET utf8mb4 COLLATE utf8mb4_unicode_ci,
    `created_by` int DEFAULT NULL,
    `created_at` timestamp NULL DEFAULT CURRENT_TIMESTAMP,
    PRIMARY KEY (`id`),
    KEY `idx_material` (`material_id`),
    KEY `idx_type` (`type`),
    KEY `idx_date` (`created_at`),
    KEY `created_by` (`created_by`),
    CONSTRAINT `movements_ibfk_1` FOREIGN KEY (`material_id`) REFERENCES
`materials` (`id`) ON DELETE CASCADE,
    CONSTRAINT `movements_ibfk_2` FOREIGN KEY (`created_by`) REFERENCES `users`
(`id`) ON DELETE SET NULL
) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4 COLLATE=utf8mb4_unicode_ci;
`

```

Tabela: notifications

```

`sql
CREATE TABLE `notifications` (
    `id` int NOT NULL AUTO_INCREMENT,
    `user_id` int NOT NULL,
    `type` enum('info','warning','error','success') CHARACTER SET utf8mb4
COLLATE utf8mb4_unicode_ci DEFAULT 'info',
    `title` varchar(200) CHARACTER SET utf8mb4 COLLATE utf8mb4_unicode_ci NOT
NULL,
    `message` text CHARACTER SET utf8mb4 COLLATE utf8mb4_unicode_ci,
    `link` varchar(500) CHARACTER SET utf8mb4 COLLATE utf8mb4_unicode_ci
DEFAULT NULL,
    `is_read` tinyint(1) NOT NULL DEFAULT '0',
    `created_at` timestamp NULL DEFAULT CURRENT_TIMESTAMP,
    PRIMARY KEY (`id`),
    KEY `idx_user_read` (`user_id`,`is_read`),
    CONSTRAINT `notifications_ibfk_1` FOREIGN KEY (`user_id`) REFERENCES
`users` (`id`) ON DELETE CASCADE
) ENGINE=InnoDB AUTO_INCREMENT=2 DEFAULT CHARSET=utf8mb4
COLLATE=utf8mb4_unicode_ci;
`

```

Tabela: organizations

```

`sql
CREATE TABLE `organizations` (
    `id` varchar(50) CHARACTER SET utf8mb4 COLLATE utf8mb4_unicode_ci NOT NULL,

```

```

    `name` varchar(200) CHARACTER SET utf8mb4 COLLATE utf8mb4_unicode_ci NOT
NULL,
    `plan` varchar(50) CHARACTER SET utf8mb4 COLLATE utf8mb4_unicode_ci DEFAULT
'Enterprise',
    `logo` longtext CHARACTER SET utf8mb4 COLLATE utf8mb4_unicode_ci,
    `policies` json DEFAULT NULL,
    `created_at` timestamp NULL DEFAULT CURRENT_TIMESTAMP,
    `updated_at` timestamp NULL DEFAULT CURRENT_TIMESTAMP ON UPDATE
CURRENT_TIMESTAMP,
    `setup_done` tinyint(1) NOT NULL DEFAULT '0',
    `owner_id` int DEFAULT NULL,
    `logo_url` varchar(500) CHARACTER SET utf8mb4 COLLATE utf8mb4_unicode_ci
DEFAULT NULL,
    `slug` varchar(100) CHARACTER SET utf8mb4 COLLATE utf8mb4_unicode_ci
DEFAULT NULL,
    `cnpj` varchar(18) CHARACTER SET utf8mb4 COLLATE utf8mb4_unicode_ci DEFAULT
NULL,
    `email` varchar(255) CHARACTER SET utf8mb4 COLLATE utf8mb4_unicode_ci
DEFAULT NULL,
    `phone` varchar(30) CHARACTER SET utf8mb4 COLLATE utf8mb4_unicode_ci
DEFAULT NULL,
    `address` text CHARACTER SET utf8mb4 COLLATE utf8mb4_unicode_ci,
    `city` varchar(100) CHARACTER SET utf8mb4 COLLATE utf8mb4_unicode_ci
DEFAULT NULL,
    `state` varchar(50) CHARACTER SET utf8mb4 COLLATE utf8mb4_unicode_ci
DEFAULT NULL,
    PRIMARY KEY (`id`)
) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4 COLLATE=utf8mb4_unicode_ci;

```

Tabela: password_resets

```

`sql
CREATE TABLE `password_resets` (
    `id` int NOT NULL AUTO_INCREMENT,
    `user_id` int NOT NULL,
    `token` varchar(255) CHARACTER SET utf8mb4 COLLATE utf8mb4_unicode_ci NOT
NULL,
    `used` tinyint(1) NOT NULL DEFAULT '0',
    `expires_at` timestamp NOT NULL,
    `created_at` timestamp NULL DEFAULT CURRENT_TIMESTAMP,
    PRIMARY KEY (`id`),
    UNIQUE KEY `token` (`token`),
    KEY `user_id` (`user_id`),
    CONSTRAINT `password_resets_ibfk_1` FOREIGN KEY (`user_id`) REFERENCES
`users` (`id`) ON DELETE CASCADE
) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4 COLLATE=utf8mb4_unicode_ci;

```

Tabela: plans

```

`sql
CREATE TABLE `plans` (
    `id` int NOT NULL AUTO_INCREMENT,
    `slug` varchar(30) CHARACTER SET utf8mb4 COLLATE utf8mb4_unicode_ci NOT
NULL,
    `name` varchar(100) CHARACTER SET utf8mb4 COLLATE utf8mb4_unicode_ci NOT
NULL,
    `price_monthly` decimal(10,2) DEFAULT '0.00',
    `price_annual` decimal(10,2) DEFAULT '0.00',
    `max_users` int DEFAULT '1',
    `max_materials` int DEFAULT '100',
    `features` json DEFAULT NULL COMMENT '["dashboard","mapa_2d","api"]',

```

```

    `sla_uptime` decimal(4,2) DEFAULT '0.00',
    `is_active` tinyint(1) NOT NULL DEFAULT '1',
    `created_at` timestamp NULL DEFAULT CURRENT_TIMESTAMP,
    PRIMARY KEY (`id`),
    UNIQUE KEY `slug` (`slug`)
) ENGINE=InnoDB AUTO_INCREMENT=4 DEFAULT CHARSET=utf8mb4
COLLATE=utf8mb4_unicode_ci;
`

```

Tabela: produto

```

`sql
CREATE TABLE `produto` (
  `codigo` int NOT NULL,
  `preco` double DEFAULT NULL,
  `descricao` varchar(255) DEFAULT NULL,
  `marca` varchar(255) DEFAULT NULL,
  PRIMARY KEY (`codigo`)
) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4 COLLATE=utf8mb4_0900_ai_ci;
`

```

Tabela: produtos

```

`sql
CREATE TABLE `produtos` (
  `id` int NOT NULL AUTO_INCREMENT,
  `nome` varchar(100) DEFAULT NULL,
  `preco` decimal(10,2) DEFAULT NULL,
  `quantidade` int DEFAULT NULL,
  PRIMARY KEY (`id`)
) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4 COLLATE=utf8mb4_0900_ai_ci;
`

```

Tabela: professor

```

`sql
CREATE TABLE `professor` (
  `codigo` int NOT NULL,
  `salario` double DEFAULT NULL,
  `nome` varchar(255) DEFAULT NULL,
  PRIMARY KEY (`codigo`)
) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4 COLLATE=utf8mb4_0900_ai_ci;
`

```

Tabela: purchase_order_items

```

`sql
CREATE TABLE `purchase_order_items` (
  `id` int NOT NULL AUTO_INCREMENT,
  `po_id` int NOT NULL,
  `material_id` int NOT NULL,
  `quantity` int NOT NULL,
  `unit_price` decimal(12,2) NOT NULL,
  `received_qty` int DEFAULT '0',
  PRIMARY KEY (`id`),
  KEY `po_id` (`po_id`),
  KEY `material_id` (`material_id`),
  CONSTRAINT `purchase_order_items_ibfk_1` FOREIGN KEY (`po_id`) REFERENCES
`purchase_orders` (`id`) ON DELETE CASCADE,
  CONSTRAINT `purchase_order_items_ibfk_2` FOREIGN KEY (`material_id`)
REFERENCES `materials` (`id`)
) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4 COLLATE=utf8mb4_unicode_ci;
`

```

Tabela: purchase_orders

```

`sql
CREATE TABLE `purchase_orders` (
  `id` int NOT NULL AUTO_INCREMENT,
  `po_number` varchar(30) CHARACTER SET utf8mb4 COLLATE utf8mb4_unicode_ci
NOT NULL,
  `supplier_id` int NOT NULL,
  `status` enum('draft','sent','confirmed','partial','received','cancelled')
CHARACTER SET utf8mb4 COLLATE utf8mb4_unicode_ci DEFAULT 'draft',
  `total_amount` decimal(12,2) DEFAULT '0.00',
  `currency` varchar(5) CHARACTER SET utf8mb4 COLLATE utf8mb4_unicode_ci
DEFAULT 'BRL',
  `expected_at` date DEFAULT NULL,
  `received_at` timestamp NULL DEFAULT NULL,
  `notes` text CHARACTER SET utf8mb4 COLLATE utf8mb4_unicode_ci,
  `created_by` int DEFAULT NULL,
  `created_at` timestamp NULL DEFAULT CURRENT_TIMESTAMP,
  `updated_at` timestamp NULL DEFAULT CURRENT_TIMESTAMP ON UPDATE
CURRENT_TIMESTAMP,
  PRIMARY KEY (`id`),
  UNIQUE KEY `po_number` (`po_number`),
  KEY `supplier_id` (`supplier_id`),
  KEY `created_by` (`created_by`),
  CONSTRAINT `purchase_orders_ibfk_1` FOREIGN KEY (`supplier_id`) REFERENCES
`suppliers` (`id`),
  CONSTRAINT `purchase_orders_ibfk_2` FOREIGN KEY (`created_by`) REFERENCES
`users` (`id`) ON DELETE SET NULL
) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4 COLLATE=utf8mb4_unicode_ci;

```

Tabela: sessions

```

`sql
CREATE TABLE `sessions` (
  `id` int NOT NULL AUTO_INCREMENT,
  `user_id` int NOT NULL,
  `token` varchar(500) CHARACTER SET utf8mb4 COLLATE utf8mb4_unicode_ci NOT
NULL,
  `provider` varchar(20) CHARACTER SET utf8mb4 COLLATE utf8mb4_unicode_ci
DEFAULT 'local',
  `ip_address` varchar(45) CHARACTER SET utf8mb4 COLLATE utf8mb4_unicode_ci
DEFAULT NULL,
  `user_agent` text CHARACTER SET utf8mb4 COLLATE utf8mb4_unicode_ci,
  `expires_at` timestamp NOT NULL,
  `created_at` timestamp NULL DEFAULT CURRENT_TIMESTAMP,
  PRIMARY KEY (`id`),
  KEY `idx_token` (`token`(255)),
  KEY `idx_user` (`user_id`),
  KEY `idx_expires` (`expires_at`),
  CONSTRAINT `sessions_ibfk_1` FOREIGN KEY (`user_id`) REFERENCES `users`
(`id`) ON DELETE CASCADE
) ENGINE=InnoDB AUTO_INCREMENT=14 DEFAULT CHARSET=utf8mb4
COLLATE=utf8mb4_unicode_ci;
`

```

Tabela: shelf_assignments

```

`sql
CREATE TABLE `shelf_assignments` (
  `id` int NOT NULL AUTO_INCREMENT,
  `shelf_id` int NOT NULL,
  `level` int NOT NULL,
  `position` int NOT NULL,
  `material_id` int NOT NULL,

```

```

    `assigned_at` timestamp NULL DEFAULT CURRENT_TIMESTAMP,
    PRIMARY KEY (`id`),
    UNIQUE KEY `uk_shelf_slot` (`shelf_id`, `level`, `position`),
    KEY `material_id` (`material_id`),
    CONSTRAINT `shelf_assignments_ibfk_1` FOREIGN KEY (`shelf_id`) REFERENCES
`shelves` (`id`) ON DELETE CASCADE,
    CONSTRAINT `shelf_assignments_ibfk_2` FOREIGN KEY (`material_id`)
REFERENCES `materials` (`id`) ON DELETE CASCADE
) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4 COLLATE=utf8mb4_unicode_ci;

```

Tabela: shelves

```

`sql
CREATE TABLE `shelves` (
  `id` int NOT NULL AUTO_INCREMENT,
  `layout_id` int NOT NULL,
  `shelf_code` varchar(5) CHARACTER SET utf8mb4 COLLATE utf8mb4_unicode_ci
NOT NULL,
  `pos_x` int NOT NULL DEFAULT '0',
  `pos_y` int NOT NULL DEFAULT '0',
  `width` int NOT NULL DEFAULT '120',
  `height` int NOT NULL DEFAULT '80',
  `levels` int NOT NULL DEFAULT '4',
  `positions` int NOT NULL DEFAULT '5',
  `created_at` timestamp NULL DEFAULT CURRENT_TIMESTAMP,
  PRIMARY KEY (`id`),
  UNIQUE KEY `uk_layout_shelf` (`layout_id`, `shelf_code`),
  CONSTRAINT `shelves_ibfk_1` FOREIGN KEY (`layout_id`) REFERENCES
`warehouse_layout` (`id`) ON DELETE CASCADE
) ENGINE=InnoDB AUTO_INCREMENT=7 DEFAULT CHARSET=utf8mb4
COLLATE=utf8mb4_unicode_ci;

```

Tabela: stock_alerts

```

`sql
CREATE TABLE `stock_alerts` (
  `id` int NOT NULL AUTO_INCREMENT,
  `material_id` int NOT NULL,
  `alert_type` enum('below_min', 'above_max', 'zero_stock', 'expiring')
CHARACTER SET utf8mb4 COLLATE utf8mb4_unicode_ci NOT NULL,
  `status` enum('active', 'acknowledged', 'resolved') CHARACTER SET utf8mb4
COLLATE utf8mb4_unicode_ci DEFAULT 'active',
  `resolved_at` timestamp NULL DEFAULT NULL,
  `resolved_by` int DEFAULT NULL,
  `created_at` timestamp NULL DEFAULT CURRENT_TIMESTAMP,
  PRIMARY KEY (`id`),
  KEY `idx_status` (`status`),
  KEY `material_id` (`material_id`),
  KEY `resolved_by` (`resolved_by`),
  CONSTRAINT `stock_alerts_ibfk_1` FOREIGN KEY (`material_id`) REFERENCES
`materials` (`id`) ON DELETE CASCADE,
  CONSTRAINT `stock_alerts_ibfk_2` FOREIGN KEY (`resolved_by`) REFERENCES
`users` (`id`) ON DELETE SET NULL
) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4 COLLATE=utf8mb4_unicode_ci;

```

Tabela: subscriptions

```

`sql
CREATE TABLE `subscriptions` (
  `id` int NOT NULL AUTO_INCREMENT,
  `user_id` int NOT NULL,

```

```

    `plan_id` int NOT NULL,
    `billing_cycle` enum('monthly','annual') CHARACTER SET utf8mb4 COLLATE
utf8mb4_unicode_ci DEFAULT 'monthly',
    `status` enum('active','past_due','cancelled','trial','expired') CHARACTER
SET utf8mb4 COLLATE utf8mb4_unicode_ci DEFAULT 'trial',
    `trial_ends_at` timestamp NULL DEFAULT NULL,
    `current_period_start` timestamp NULL DEFAULT NULL,
    `current_period_end` timestamp NULL DEFAULT NULL,
    `cancelled_at` timestamp NULL DEFAULT NULL,
    `created_at` timestamp NULL DEFAULT CURRENT_TIMESTAMP,
    `updated_at` timestamp NULL DEFAULT CURRENT_TIMESTAMP ON UPDATE
CURRENT_TIMESTAMP,
    PRIMARY KEY (`id`),
    KEY `idx_user` (`user_id`),
    KEY `idx_status` (`status`),
    KEY `plan_id` (`plan_id`),
    CONSTRAINT `subscriptions_ibfk_1` FOREIGN KEY (`user_id`) REFERENCES
`users` (`id`) ON DELETE CASCADE,
    CONSTRAINT `subscriptions_ibfk_2` FOREIGN KEY (`plan_id`) REFERENCES
`plans` (`id`)
) ENGINE=InnoDB AUTO_INCREMENT=2 DEFAULT CHARSET=utf8mb4
COLLATE=utf8mb4_unicode_ci;
`

```

Tabela: suppliers

```

`sql
CREATE TABLE `suppliers` (
    `id` int NOT NULL AUTO_INCREMENT,
    `name` varchar(200) CHARACTER SET utf8mb4 COLLATE utf8mb4_unicode_ci NOT
NULL,
    `cnpj` varchar(18) CHARACTER SET utf8mb4 COLLATE utf8mb4_unicode_ci DEFAULT
NULL,
    `contact_name` varchar(100) CHARACTER SET utf8mb4 COLLATE
utf8mb4_unicode_ci DEFAULT NULL,
    `contact_email` varchar(255) CHARACTER SET utf8mb4 COLLATE
utf8mb4_unicode_ci DEFAULT NULL,
    `contact_phone` varchar(30) CHARACTER SET utf8mb4 COLLATE
utf8mb4_unicode_ci DEFAULT NULL,
    `address` text CHARACTER SET utf8mb4 COLLATE utf8mb4_unicode_ci,
    `notes` text CHARACTER SET utf8mb4 COLLATE utf8mb4_unicode_ci,
    `is_active` tinyint(1) NOT NULL DEFAULT '1',
    `created_at` timestamp NULL DEFAULT CURRENT_TIMESTAMP,
    `updated_at` timestamp NULL DEFAULT CURRENT_TIMESTAMP ON UPDATE
CURRENT_TIMESTAMP,
    PRIMARY KEY (`id`),
    UNIQUE KEY `cnpj` (`cnpj`),
    KEY `idx_cnpj` (`cnpj`)
) ENGINE=InnoDB AUTO_INCREMENT=4 DEFAULT CHARSET=utf8mb4
COLLATE=utf8mb4_unicode_ci;
`

```

Tabela: system_settings

```

`sql
CREATE TABLE `system_settings` (
    `id` int NOT NULL AUTO_INCREMENT,
    `setting_key` varchar(100) CHARACTER SET utf8mb4 COLLATE utf8mb4_unicode_ci
NOT NULL,
    `setting_val` text CHARACTER SET utf8mb4 COLLATE utf8mb4_unicode_ci,
    `description` varchar(300) CHARACTER SET utf8mb4 COLLATE utf8mb4_unicode_ci
DEFAULT NULL,
    `updated_at` timestamp NULL DEFAULT CURRENT_TIMESTAMP ON UPDATE
CURRENT_TIMESTAMP,

```

```

    PRIMARY KEY (`id`),
    UNIQUE KEY `setting_key` (`setting_key`)
) ENGINE=InnoDB AUTO_INCREMENT=11 DEFAULT CHARSET=utf8mb4
COLLATE=utf8mb4_unicode_ci;
`

```

Tabela: users

```

`sql
CREATE TABLE `users` (
  `id` int NOT NULL AUTO_INCREMENT,
  `name` varchar(100) CHARACTER SET utf8mb4 COLLATE utf8mb4_unicode_ci NOT
NULL,
  `surname` varchar(100) CHARACTER SET utf8mb4 COLLATE utf8mb4_unicode_ci
DEFAULT NULL,
  `email` varchar(255) CHARACTER SET utf8mb4 COLLATE utf8mb4_unicode_ci NOT
NULL,
  `password_hash` varchar(255) CHARACTER SET utf8mb4 COLLATE
utf8mb4_unicode_ci DEFAULT NULL,
  `company` varchar(200) CHARACTER SET utf8mb4 COLLATE utf8mb4_unicode_ci
DEFAULT NULL,
  `phone` varchar(30) CHARACTER SET utf8mb4 COLLATE utf8mb4_unicode_ci
DEFAULT NULL,
  `role` enum('superadmin','admin','manager','operator','viewer') CHARACTER
SET utf8mb4 COLLATE utf8mb4_unicode_ci NOT NULL DEFAULT 'viewer',
  `avatar_url` longtext CHARACTER SET utf8mb4 COLLATE utf8mb4_unicode_ci,
  `org_id` varchar(50) CHARACTER SET utf8mb4 COLLATE utf8mb4_unicode_ci
DEFAULT 'org_1',
  `sector` varchar(100) CHARACTER SET utf8mb4 COLLATE utf8mb4_unicode_ci
DEFAULT 'Logística',
  `provider` enum('local','google','microsoft','webauthn') CHARACTER SET
utf8mb4 COLLATE utf8mb4_unicode_ci DEFAULT 'local',
  `provider_id` varchar(255) CHARACTER SET utf8mb4 COLLATE utf8mb4_unicode_ci
DEFAULT NULL,
  `is_active` tinyint(1) NOT NULL DEFAULT '1',
  `email_verified` tinyint(1) NOT NULL DEFAULT '0',
  `last_login_at` timestamp NULL DEFAULT NULL,
  `failed_logins` int NOT NULL DEFAULT '0',
  `locked_until` timestamp NULL DEFAULT NULL,
  `created_at` timestamp NULL DEFAULT CURRENT_TIMESTAMP,
  `updated_at` timestamp NULL DEFAULT CURRENT_TIMESTAMP ON UPDATE
CURRENT_TIMESTAMP,
  `setup_done` tinyint(1) NOT NULL DEFAULT '0',
  `position_label` varchar(100) CHARACTER SET utf8mb4 COLLATE
utf8mb4_unicode_ci DEFAULT NULL,
  `position_id` int DEFAULT NULL,
  `dob` varchar(50) COLLATE utf8mb4_unicode_ci DEFAULT NULL,
  `gender` varchar(50) COLLATE utf8mb4_unicode_ci DEFAULT NULL,
  `country` varchar(100) COLLATE utf8mb4_unicode_ci DEFAULT NULL,
  PRIMARY KEY (`id`),
  UNIQUE KEY `email` (`email`),
  KEY `idx_email` (`email`),
  KEY `idx_role` (`role`),
  KEY `idx_provider` (`provider`),
  KEY `org_id` (`org_id`),
  CONSTRAINT `users_ibfk_1` FOREIGN KEY (`org_id`) REFERENCES `organizations`
(`id`) ON DELETE SET NULL
) ENGINE=InnoDB AUTO_INCREMENT=5 DEFAULT CHARSET=utf8mb4
COLLATE=utf8mb4_unicode_ci;
`

```

Tabela: usuarios

```
`sql
CREATE TABLE `usuarios` (
  `id` int NOT NULL AUTO_INCREMENT,
  `nome` varchar(100) DEFAULT NULL,
  `email` varchar(150) DEFAULT NULL,
  `senha` varchar(25) DEFAULT NULL,
  `perfil` varchar(20) DEFAULT NULL,
  PRIMARY KEY (`id`)
) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4 COLLATE=utf8mb4_0900_ai_ci;
```

Tabela: warehouse_layout

```
`sql
CREATE TABLE `warehouse_layout` (
  `id` int NOT NULL AUTO_INCREMENT,
  `name` varchar(100) CHARACTER SET utf8mb4 COLLATE utf8mb4_unicode_ci
  DEFAULT 'Armazém Principal',
  `room_w` int NOT NULL DEFAULT '720',
  `room_h` int NOT NULL DEFAULT '480',
  `door_x` int DEFAULT '620',
  `door_y` int DEFAULT '480',
  `door_w` int DEFAULT '100',
  `aisles` json DEFAULT NULL,
  `is_active` tinyint(1) NOT NULL DEFAULT '1',
  `created_at` timestamp NULL DEFAULT CURRENT_TIMESTAMP,
  `updated_at` timestamp NULL DEFAULT CURRENT_TIMESTAMP ON UPDATE
  CURRENT_TIMESTAMP,
  `org_id` varchar(50) CHARACTER SET utf8mb4 COLLATE utf8mb4_unicode_ci
  DEFAULT NULL,
  `blueprint_data` longtext CHARACTER SET utf8mb4 COLLATE utf8mb4_unicode_ci,
  PRIMARY KEY (`id`)
) ENGINE=InnoDB AUTO_INCREMENT=6 DEFAULT CHARSET=utf8mb4
COLLATE=utf8mb4_unicode_ci;
```

Tabela: warehouses

```
`sql
CREATE TABLE `warehouses` (
  `id` int NOT NULL AUTO_INCREMENT,
  `org_id` varchar(50) CHARACTER SET utf8mb4 COLLATE utf8mb4_unicode_ci NOT
  NULL,
  `name` varchar(200) CHARACTER SET utf8mb4 COLLATE utf8mb4_unicode_ci NOT
  NULL,
  `tab_number` int NOT NULL DEFAULT '1',
  `is_shared` tinyint(1) NOT NULL DEFAULT '0',
  `description` text CHARACTER SET utf8mb4 COLLATE utf8mb4_unicode_ci,
  `created_at` timestamp NULL DEFAULT CURRENT_TIMESTAMP,
  PRIMARY KEY (`id`),
  KEY `idx_org` (`org_id`)
) ENGINE=InnoDB AUTO_INCREMENT=2 DEFAULT CHARSET=utf8mb4
COLLATE=utf8mb4_unicode_ci;
```

Tabela: webauthn_credentials

```
`sql
CREATE TABLE `webauthn_credentials` (
  `id` int NOT NULL AUTO_INCREMENT,
  `user_id` int NOT NULL,
  `credential_id` text CHARACTER SET utf8mb4 COLLATE utf8mb4_unicode_ci NOT
  NULL,
  `public_key` text CHARACTER SET utf8mb4 COLLATE utf8mb4_unicode_ci NOT NULL,
```

```
`sign_count` int DEFAULT '0',
`device_name` varchar(100) CHARACTER SET utf8mb4 COLLATE utf8mb4_unicode_ci
DEFAULT NULL,
`created_at` timestamp NULL DEFAULT CURRENT_TIMESTAMP,
PRIMARY KEY (`id`),
KEY `user_id` (`user_id`),
CONSTRAINT `webauthn_credentials_ibfk_1` FOREIGN KEY (`user_id`) REFERENCES
`users` (`id`) ON DELETE CASCADE
) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4 COLLATE=utf8mb4_unicode_ci;
```